

End-to-End Data Session Flow

Components Involved:

- User Equipment (UE): Smartphone, tablet, IoT device that accesses the network
- Next Generation Node B (gNB): 5G base station providing radio connectivity
- Access and Mobility (AMF): Handles registration, authentication, and mobility
- Session Management Function (SMF): Creates and manages PDU sessions
- User Plane Function (UPF): Routes user data packets to and from the internet
- Data Network (Internet): External network providing web services and applications

Phase 1: Registration

Registration is the first procedure performed when a user device connects to the 5G network. During this phase, the user is authenticated and allowed to access network services.

Registration Steps:

Step 1: The UE powers on and searches for nearby 5G cells

Step 2: The UE sends a Registration Request to the gNB

Step 3: The gNB forwards the request to the AMF

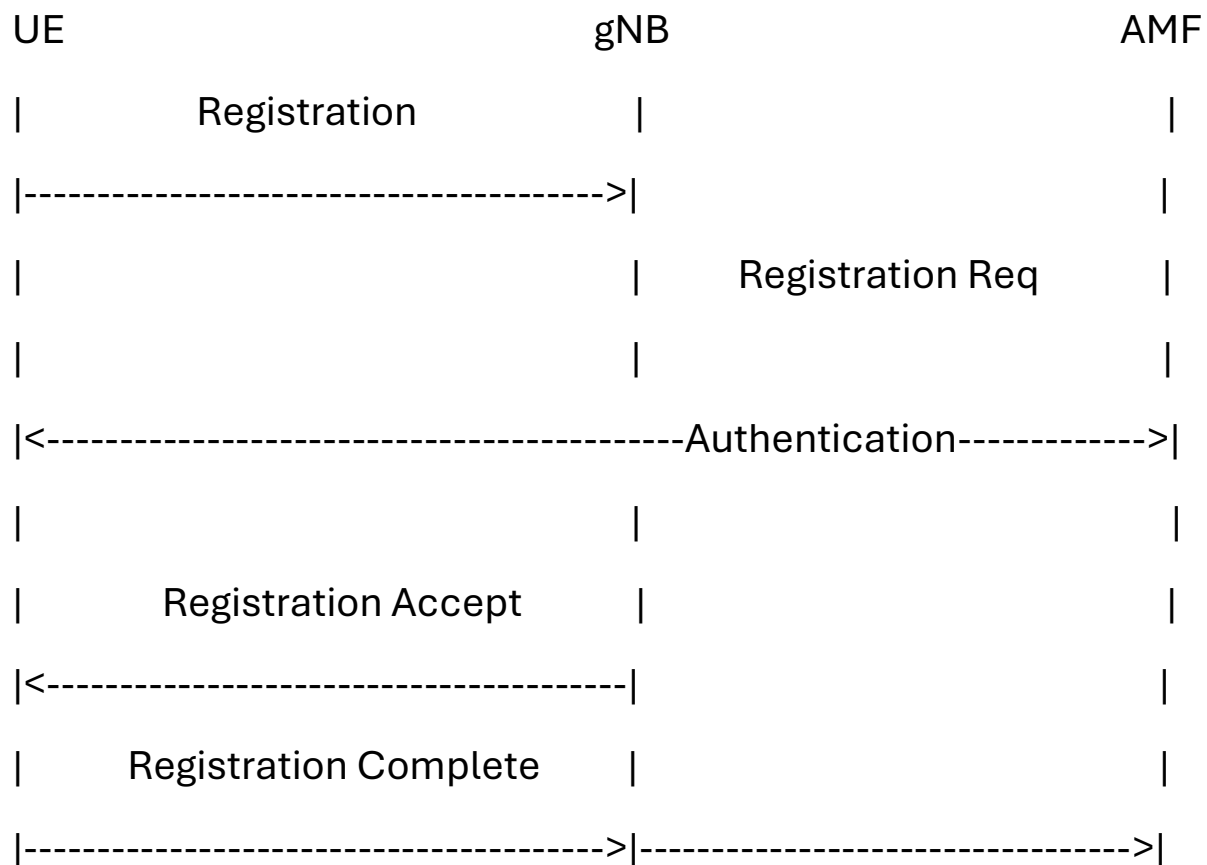
Step 4: The AMF authenticates the user using subscriber information.

Authentication includes:

- SIM verification
- Identity verification
- Security key generation

Step 5: The AMF sends a Registration Accept message back to the UE

Registration Sequence Diagram



Phase 2: PDU Session Establishment

A Protocol Data Unit (PDU) Session is the logical connection that allows a user to access external networks such as the Inter

Without a PDU Session, the user cannot send or receive internet data.

PDU Session Establishment Steps:

Step 1: The UE requests internet access

Step 2: The request reached the AMF

Step 3: The AMF forwards the request to the SMF

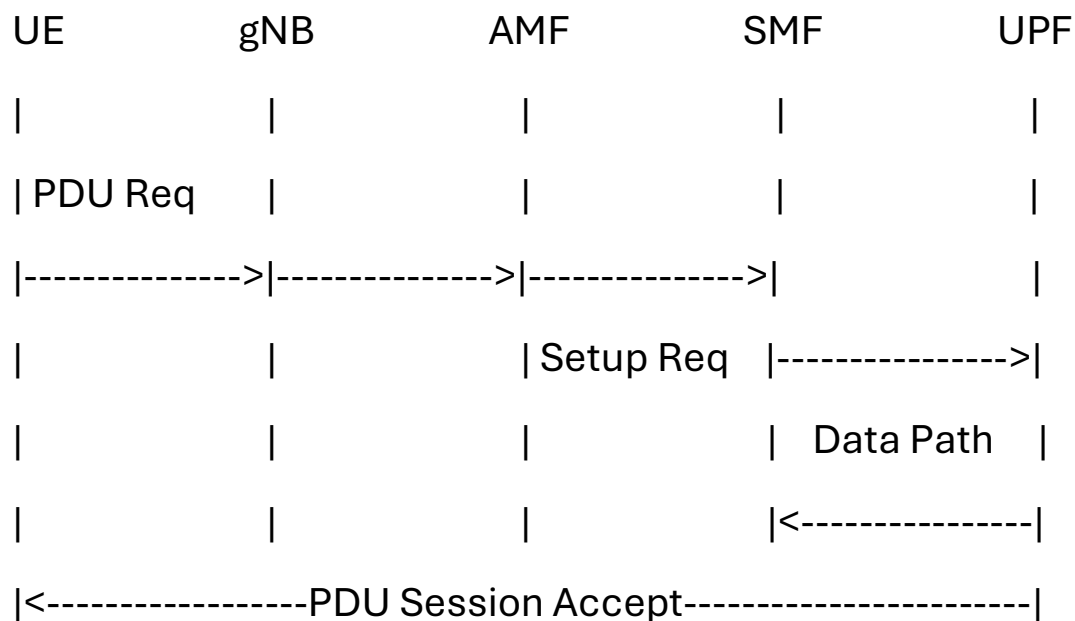
Step 4: The SMF selects an appropriate UPF

Step 5: The SMF establishes the PDU Session

Step 6: The UPF allocates the user data path and IP address.

Step 7: The UE receives confirmation that the session is active

PDU Session Sequence Diagram



Phase 3: User Plane Traffic Flow

Once the PDU Session is established, the actual user data begins flowing through the network.

Examples include:

- Web browsing
- Video streaming
- Voice over 5G

- Online gaming
- File downloads

User Plane Traffic Diagram:

UE

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| User Data

gNB

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UPF

|

Internet

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Cloud Services

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UPF

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gNB

|

UE

After the PDU Session is established, user data travels directly through the UPF. The AMF and SMF remain responsible for control and session management but do not carry user traffic.